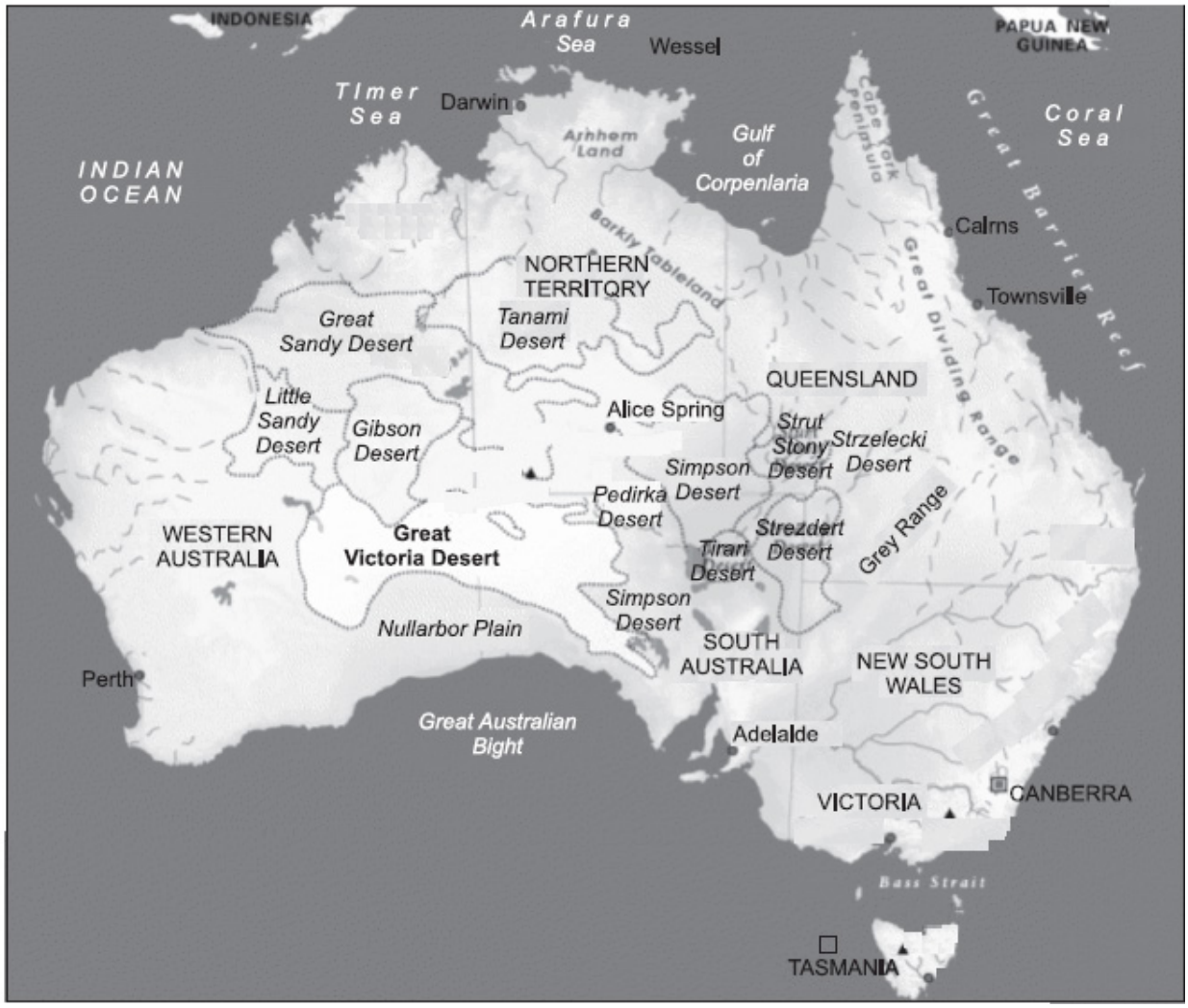
Explanation (66)



69. The Southern Annular Mode can result in enhanced rainfall in regions of southern Australia.

What is it?

The Southern Annular Mode, or SAM, also known as the Antarctic Oscillation (AAO), is a mode of variability which can affect rainfall in southern Australia. The SAM refers to the north/south movement of the strong westerly winds that dominate the middle to higher latitudes of the Southern Hemisphere. The belt of strong westerly winds in the Southern Hemisphere is also associated with the storm systems and cold fronts that move from west to east.

During a “positive” SAM event, the belt of strong westerly winds contracts towards the south pole. This results in weaker than normal westerly winds and higher pressure over southern